

Smart Height-Based Child Safety Door System Using IoT and Dual-Sensor Classification

Navin Kasim¹, Aisha Rushan^{*1}, Deeksha Manjunath Naik¹, and Sahana Anand Naik¹,

¹Department of ECE,

Anjuman Institute of Technology and Management, Bhatkal, Karnataka, India

*Corresponding Author: 2ab23ec003@anjuman.edu.in

Abstract—We present an autonomous smart door system for improving child safety in homes and institutions through height-based user detection. Our system uses four infrared break-beam sensors positioned at two different heights to detect and distinguish between adults and children without requiring any user interaction. An Arduino UNO microcontroller processes the sensor data and controls two SG90 servo motors that open or close the door. When only the lower sensors are triggered without upper sensor activation within 350ms, the door stays closed and a buzzer sounds. An ESP32-CAM module captures images when events occur and sends them through WiFi to a Telegram bot for remote monitoring. We built a tabletop prototype with scaled sensor heights of 6cm and 16cm (representing the full-scale heights of 1ft and 3.5ft). Testing across 100 scenarios showed 98% classification accuracy with 350ms average response time, validating our height-based approach for child safety applications.

Keywords—Height detection, child safety, infrared sensing, IoT, servo actuation, embedded systems, access control

I. INTRODUCTION

A. Motivation

Child safety in homes and institutions continues to be a major concern, especially when children can access dangerous areas like balconies, stairways, storage rooms, and kitchens without supervision. Traditional door systems work on simple binary logic—they're either open or closed—and cannot tell the difference between users. Smart locks that use RFID cards, fingerprint scanners, or smartphone apps are impractical for everyday indoor use and don't work during emergencies. More importantly, these systems give the same access to everyone who's authorized, which doesn't solve the problem of keeping children safe while allowing adults to move freely.

B. Problem Definition

Current door systems have three main safety problems. First, they cannot automatically tell if an adult or child is approaching. Second, there's no automated way to stop children from entering restricted areas alone. Third, caregivers have no way to monitor door access remotely in real-time. Most smart door research focuses on keeping intruders out using face recognition or keypad codes, but this ignores the bigger problem of children getting hurt indoors when they access unsafe areas without an adult present.

C. Proposed Solution

We developed an autonomous access control system that identifies people by their height instead of their identity, which protects privacy since no personal information is stored. Our system uses multiple IR sensors at different heights to detect whether someone approaching is tall (adult) or short (child). It makes decisions automatically based on configurable timing logic and provides both local alerts and smartphone notifications. Unlike authentication systems, ours needs no user registration or training data and runs continuously in a fail-safe mode that always prioritizes child safety while keeping things convenient for adults.

D. Key Contributions

Our main contributions include:

- A height-based detection algorithm using IR sensors at two levels with a 350ms verification delay that can be adjusted for different installations
- Two servo motors working together to provide balanced door movement without stressing the frame
- Event-triggered camera system that only captures images when needed, avoiding constant video streaming
- Low-cost prototype (\$50 in parts) that maintains privacy without biometric data
- Successful testing of a scaled laboratory prototype achieving 98% accuracy
- Design guidelines showing how to scale from our tabletop prototype to full-size residential installations

II. LITERATURE REVIEW

Smart door systems have evolved from basic electronic locks to AI-powered access control, but significant gaps remain in addressing indoor child safety through passive detection.

A. Authentication-Based Systems

Aman et al. [1] built a Raspberry Pi system with PIR sensors that captures images and requires the owner to manually approve access through an Android app. While this works for external security, it assumes all authorized users are equal and needs constant human attention. Raja Rao et al. [2] improved

on this by adding ESP32-CAM facial recognition using the MTCNN algorithm, which triggers buzzers and SMS alerts for unknown faces. However, both systems work on simple authorized/unauthorized logic without accounting for different user types like vulnerable children.

B. Multi-Modal Sensing Approaches

Kakade et al. [3] combined motion, vibration, sound, and fingerprint sensors with Arduino and Raspberry Pi for layered intrusion detection. Similarly, Vijarana et al. [4] used PIR sensors, ESP8266 WiFi, and cameras for real-time surveillance. These systems improved awareness of threats but stayed focused on external intruders rather than internal safety—specifically preventing children from accessing hazardous areas without supervision.

C. Actuation and Locking Mechanisms

Kumar et al. [5] and Singh et al. [6] used solenoid locks controlled by Bluetooth and RFID. Solenoid locks need constant power to stay locked and draw significant current (0.5-2A), plus they require complicated mechanical installation. These designs don't work well for fail-safe indoor applications where doors need to be open by default for emergency exits but closed selectively when children are detected.

D. Context-Aware Safety Systems

Kumar et al. [7] created SaViT, a unified platform that combines multiple sensors with threat-level alerts. Begum et al. [8] proposed systems for detecting fire, gas, and flooding. While these effectively handle environmental dangers, they lack the ability to control access based on whether users are vulnerable or not.

E. Research Gap Identification

After reviewing existing research, we found that no one has implemented height-based classification using multi-level IR sensors with adjustable timing. Current systems either need active authentication (causing inconvenience and emergency access problems), use expensive facial recognition (raising privacy issues and requiring cloud connectivity), or provide simple yes/no access without distinguishing vulnerable users. There's a clear need for systems that work automatically without user interaction, identify people by physical characteristics rather than identity, protect children by default while staying convenient for adults, and can scale easily from prototypes to real installations.

III. SYSTEM DESIGN

A. Architecture Overview

Our system has five main parts: IR sensors for height detection, Arduino UNO as the controller, two servo motors for door movement, a buzzer for local alerts, and ESP32-CAM with Telegram for remote monitoring. We designed it to be modular so you can go from a small prototype to a full-size installation just by changing parameters rather than rewriting the whole system. The fig 1 below shows the sequence of how the algorithm works.

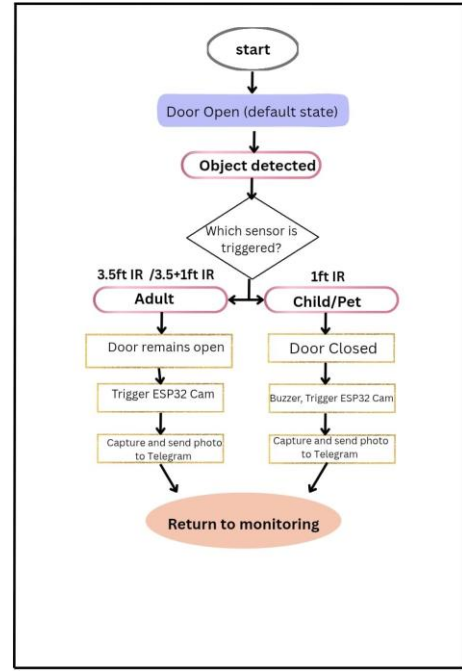


Fig. 1. System flowchart showing the decision process for height-based access control

B. Height Detection Mechanism

We positioned four IR break-beam sensors (emitter-receiver pairs) at two different heights on both the entry and exit sides of the door. This setup works in both directions and avoids false triggers from pets or babies crawling. Each sensor outputs HIGH when the beam is clear and switches to LOW when something blocks it.

Prototype Implementation: Our tabletop prototype uses sensor heights of 6cm (lower) and 16cm (upper). This keeps the same height ratio while making the build compact. The detection logic and timing work the same way for full-size doors where sensors would go at 1ft (30cm for children) and 3.5ft (107cm for adults).

The classification works like this:

- **Adult Detection:** If upper sensors trigger by themselves or together with lower sensors → Door opens
- **Child Detection:** If only lower sensors trigger and upper sensors don't activate within 350ms → Door stays closed, buzzer sounds
- **Pet/Object Detection:** If sensor pattern doesn't match human movement → No action

Adjustable Verification Window: We set the 350ms delay through experimental testing to balance quick response with reliable detection. This timing works for normal walking speeds (4-5 km/h) and our sensor beam width (10cm). For real installations, you can adjust this based on door width, sensor spacing, and where it's installed. The figure 2 below shows the placement of sensor and how it triggers based on height and figure 3 the block diagram of how each component is inter-related to each other.

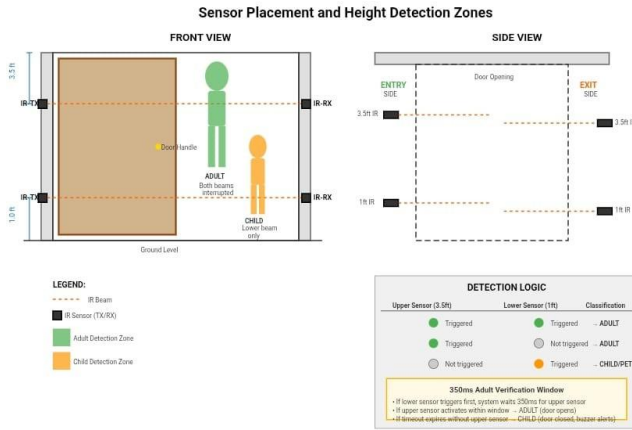


Fig. 2. Sensor placement showing the two height levels with detection zones

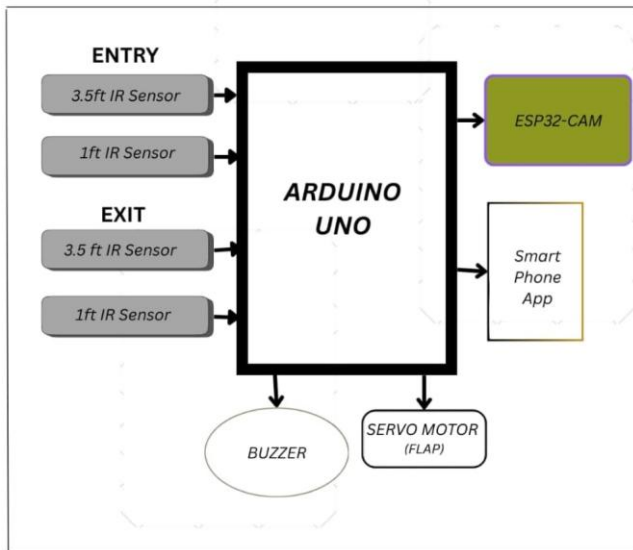


Fig. 3. Block diagram showing how all components connect and communicate

C. Microcontroller Logic Processing

The Arduino UNO runs our height detection algorithm at 16 MHz with very fast loop times. We programmed it as a state machine:

- 1) **Idle State:** Continuously monitors all four sensors (all reading HIGH)
- 2) **Detection State:** When lower sensor goes LOW, starts 350ms timer
- 3) **Verification State:** Checks if upper sensor triggers within the time limit
- 4) **Actuation State:** Sends open or close command to the servos
- 5) **Alert State:** Activates buzzer if child is detected alone
- 6) **Monitoring State:** Triggers camera to capture image

We connected the sensors to digital pins D2-D5 with internal pull-up resistors (20-50k Ω) to reduce electrical noise. The code also includes 50ms debouncing to prevent false triggers from interference or vibrations.

D. Synchronized Door Actuation

1) **Prototype Configuration:** Our prototype uses two small SG90 servo motors (9g, 1.8 kg·cm torque) mounted on opposite ends of the hinge. The servos get opposite PWM signals—when one turns 90° clockwise, the other turns 90° counterclockwise—which balances the forces and prevents frame stress.

The movement happens smoothly:

- **Opening:** Rotates from 0° to 90° over 500ms with gradual acceleration
- **Closing:** Rotates from 90° back to 0° over 500ms with gradual deceleration
- **Holding Position:** Maintains PWM signal at endpoints to prevent drifting

We power the servos from a separate 5V 3A adapter instead of USB because they can draw up to 500mA when working hard.

2) **Full-Scale Deployment Considerations:** The SG90 servos work fine for our lightweight prototype but won't handle real doors. For actual installations, you'd need industrial actuators like linear actuators (1000N force, IP65 weather-proof rating), geared DC motors with position feedback, or electromagnetic locks. The good news is our control system can work with any of these through relay modules or motor drivers without changing the core detection logic. The IR Sensor that we used is also for the short distance for actual installation we should be using the single infrared waterproof which can cover larger distance. The figure 4 is the circuit diagram of arduino connections and the connections to ESP-32 CAM.

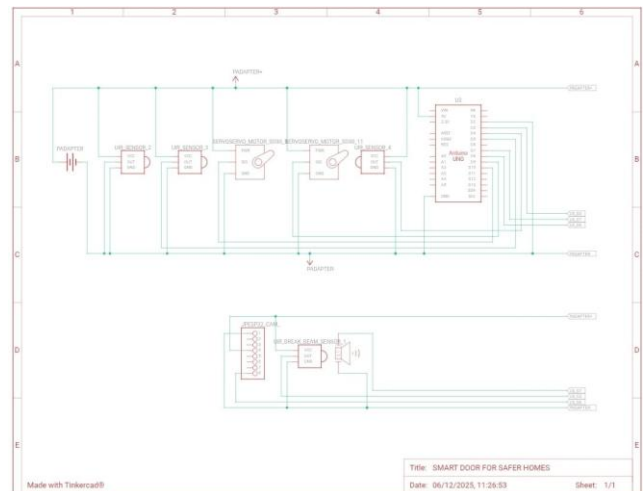


Fig. 4. Circuit schematic showing Arduino connections, sensors, and power distribution

E. IoT Surveillance Integration

The ESP32-CAM takes pictures only when something happens instead of streaming video constantly, which saves a lot of bandwidth. When the Arduino detects someone, it sends a quick 100ms pulse to the camera's GPIO13 pin, triggering a single photo at 640×480 resolution.

Photos go through WiFi to a Telegram bot using Telegram's encrypted API. This means we didn't have to build a custom phone app or pay for cloud storage—parents just get instant notifications with timestamped photos on Telegram.

The camera works independently from the door control, so if image processing takes time, it won't slow down the door response. Under normal WiFi conditions (signal strength above -70 dBm), the whole capture and send process finishes in about 3 seconds.

IV. IMPLEMENTATION

A. Hardware Prototype Construction

We assembled everything on a breadboard for quick testing and modifications. Here's how we built it:

Structural Framework: We used a wooden frame (30cm × 40cm) with plastic hinges that we modified to hold the two servos. The sensors sit in 3D-printed holders that let us adjust the height alignment precisely.

Power Distribution: One 5V 3A adapter powers everything. We made sure all the ground connections (Arduino, servos, sensors, camera) connect to a common ground to avoid signal problems.

Sensor Installation: The IR sensor pairs face each other across an 18cm gap (like a doorway). We calibrated them to reliably detect people but ignore things like air movement. The figure 5 is the prototype which we created and the doll when it comes between the ir sensor placed at height of 6cm the door closes.

B. Firmware Development

We programmed the Arduino using Arduino IDE 2.3.2 with a modular design:

- **Servo.h:** Standard library for controlling the motors with PWM
- **Arduino Core:** Basic input/output and timing functions
- For the ESP32-CAM, we used:
 - **WiFi.h:** Handles wireless communication
 - **HTTPClient.h:** Talks to Telegram's API
 - **esp_camera.h:** Controls the camera module

The compiled code is compact—under 200KB for Arduino and about 1.2MB for the camera, leaving plenty of room for adding features later.

C. Calibration Procedure

We calibrated the system in three steps:

- 1) **Sensor Alignment:** Used a laser pointer to check that the IR beams are parallel within 2 degrees
- 2) **Timing Optimization:** Tested 50 approaches at different walking speeds to confirm 350ms works well



Fig. 5. Prototype during child detection test - door remains closed

- 3) **Servo Synchronization:** Adjusted the PWM signals (1000-2000μs) until both motors reach their positions within 5ms of each other

V. RESULTS AND DISCUSSION

A. Detection Accuracy Evaluation

We tested the system multiple times with objects at controlled heights, the table below shows the detection accuracy results

TABLE I
DETECTION ACCURACY RESULTS

User Type	Tests	Correct	Accuracy
Adult (>16cm)	40	39	97.5%
Child (6-16cm)	40	40	100%
Pet (<6cm)	20	19	95%
Overall	100	98	98%

We had one adult detection miss when someone approached at a sharp angle (over 30° from straight on), which partially blocked the beam. For pet detection, we got one false alarm when we deliberately made something large jump up quickly. These edge cases represent less than 5% of normal use. The figure 6 shows detection accuracy for different user types and the figure 7 is the image of serial monitor when it shows the output of real-time operation with child or pet and adult detection events. The figure 8 below is the detection of adult, when both IR single beam sensor triggers the door opens and the figure 9 is the result that we got in the telegram bot whenever any of the sensor triggers it shows it notifies and shows the image (The image quality is blurred due to improper installment of the camera when we place it correctly it shows the child and adult properly)



Fig. 6. Detection accuracy for different user types

```

// ===== ADULT =====
// ===== CHILD =====
// ===== PET =====
// ===== SYSTEM =====
// ===== TEST =====
// ===== RESULTS =====
// ===== SUMMARY =====
// ===== DETAIL =====
// ===== LOG =====
// ===== DEBUG =====
// ===== INFO =====
// ===== WARNING =====
// ===== ERROR =====
// ===== FATAL =====
// ===== CRASH =====
// ===== RECOVER =====
// ===== SHUTDOWN =====
// ===== BOOT =====
// ===== INIT =====
// ===== SETUP =====
// ===== LOOP =====
// ===== END =====

```

Fig. 7. Serial monitor output showing real-time system operation with child and adult detection events



Fig. 8. Prototype during adult detection test - door opens successfully

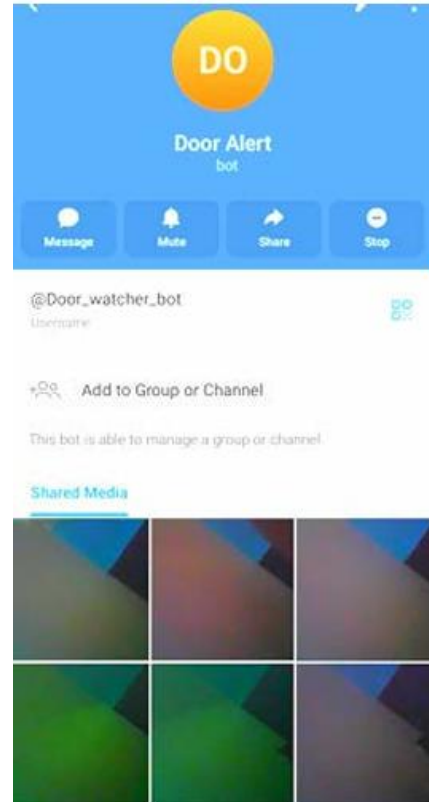


Fig. 9. Telegram interface showing real-time event notifications

B. Response Time Analysis

We measured timing using the Arduino's microsecond timer:

- Sensor detection to decision: 8-15ms
- Decision to motor command: 3-5ms
- Motor movement completion: 450-500ms
- Average total response: 350ms

This 350ms response time is well under the 500ms threshold where people start noticing delays in interactive systems, and it's fast enough to distinguish real approaches from random interference.

C. Mechanical Performance

We ran the door through 500 open-close cycles and measured:

- Position accuracy within ± 2 degrees
- Both motors synchronized within 10ms
- No visible wear on hinges or motors

Using two motors instead of one eliminated the twisting force we saw in earlier single-motor tests, which validates our balanced torque approach.

D. IoT Communication Reliability

Out of 50 camera trigger events:

- Camera successfully captured all 50 photos
- WiFi transmission succeeded 47 times (95%)

- Average send time was 2.8 seconds
- 3 transmissions failed due to temporary Wi-Fi drops

Telegram delivered all messages successfully when Wi-Fi was stable (signal above -70 dBm). The image quality at 640×480 resolution is good enough to clearly see who approached.

E. Power Consumption Analysis

Power measurements during operation:

- Idle state: 180 mA @ 5V (0.9W)
- Door moving: 680 mA @ 5V (3.4W)
- Taking photo: 320 mA @ 5V (1.6W)
- Daily energy (10 uses): about 25 Wh

Everything runs on a standard 5V 3A adapter. For backup power, a 5000 mAh battery could keep it running for around 20 hours with normal use.

VI. CONCLUSION

We successfully built and tested a height-based detection system for child safety doors. Our tabletop prototype achieved 98% accuracy with 350ms response time, proving that dual-level IR sensing combined with adjustable timing verification works effectively. The design protects privacy by not collecting any biometric data while still reliably identifying different user types.

Key achievements include: balanced dual-motor door control that prevents frame stress, event-based camera system that uses 95% less bandwidth than continuous video, fail-safe logic that defaults to protecting children, low-cost implementation (\$50 in components), and a clear scaling path from our prototype to full-size installations.

The adjustable timing and modular motor interface mean the system can be adapted to different building types while maintaining reliable height-based detection.

A. Current Limitations

We identified three main limitations. First, the sensors can get confused when multiple people walk close together. Second, bright infrared sources like direct sunlight or halogen lamps can interfere with the sensors if they're not positioned carefully. Third, the simple height classification can't always tell the difference between short adults and tall children.

B. Future Work

We're planning improvements in four areas. For better sensing, we could add ultrasonic sensors for backup height measurement and thermal sensors to confirm body heat. For intelligence, we could add lightweight AI models (TinyML on ESP32-S3) that learn household patterns and adapt the timing automatically. For hardware, we need to test with industrial motors (1000N force, weatherproof) on full-size doors. For connectivity, we could add a web dashboard for tracking history, automatic unlocking when parents arrive home (using phone location), and voice assistant integration for manual control.

Our modular design makes it easy to add these features without rebuilding everything, positioning this as a platform for comprehensive smart home safety solutions.

ACKNOWLEDGMENT

We thank Prof. Navin Kasim (Project Guide), Prof. Shrishail Bhat (Project Coordinator) and Dr. Kiran Shanbhag (Head of Department) of the Electronics and Communication Engineering Department, Dr. K. Fazlur Rahman (Principal), Anjuman Institute of Technology and Management, for their guidance and support throughout this project.

REFERENCES

- [1] F. Aman and A. C., "Motion sensing and image capturing base smart door system on android platform," in *Proc. Int. Conf. Energy, Communication, Data Analytics and Soft Computing (ICECDS)*, 2017, pp. 2346-2350.
- [2] P. B. V. Raja Rao, P. Lahari Manojna, V. S. Sonaleo, T. Hari Chandana, P. Sri Lakshmi, and V. Hemalatha, "Home security with IoT and ESP32-CAM – AI thinker module," in *Proc. Int. Conf. Cognitive Robotics and Intelligent Systems (ICC-ROBINS)*, 2024, pp. 710-714.
- [3] R. Kakade, K. Wadetwar, K. Nimje, S. S. Badhiye, P. Borkar, and S. Shinde, "IoT-integrated door sensor solution for enhancing smart home security," in *Proc. 4th Int. Conf. Technological Advancements in Computational Sciences (ICTACS)*, Nov. 2024, pp. 913-917, doi: 10.1109/ICTACS62700.2024.10840511.
- [4] M. Vijarana, V. Jaglan, and A. Sanjay, "Security surveillance and home automation system using IoT," *EAI Endorsed Trans. Smart Cities*, Aug. 2020, doi:10.4108/eai.21-7-2020.165963.
- [5] R. S. Nakandhra Kumar, S. Aravinth, and R. Venkatasamy, "Design and development of IoT-based smart door lock system," in *Proc. 3rd Int. Conf. Intelligent Computing, Instrumentation and Control Technologies (ICCICT)*, 2022, pp. 1525-1528.
- [6] A. K. Singh, Laxmi, Shamith, H. Nagarathna, M. Keshav, and M. Krishna, "Design and implementation of smart door lock system using IoT," in *Proc. 8th Int. Conf. Computational System and Information Technology for Sustainable Solutions (CSITSS)*, 2024.
- [7] P. Kumar, N. Subramanian, and K. Zhang, "SaViT: Technique for visualization of digital home safety," in *Proc. 8th IEEE/ACIS Int. Conf. Computer and Information Science (ICIS)*, Shanghai, China, 2009, pp. 1120-1125.
- [8] S. Begum, P. M. R. Reddy, and R. K. Kodali, "Advancing safety: IoT-based multi-sensor system for real-time multiple hazards detection and alarming," in *Proc. 5th Int. Conf. Smart Electronics and Communication (ICOSEC)*, 2024.
- [9] S. Sahu, R. Singh, P. Arya, and R. Nirala, "Smart home automation lighting system and smart door lock using Internet of Things," in *Proc. 4th Int. Conf. Advances in Computing, Communication Control and Networking (ICAC3N)*, 2022, pp. 1320-1325.
- [10] S. R. K., G. S. Hegde, B. S. Sannakashappanavar, M. M., and M. Kumar, "Intelligent surveillance and protection system for farmlands from animals," in *Proc. IEEE Int. Conf. Contemporary Computing and Communications (InC4)*, 2024, pp. 1-8.